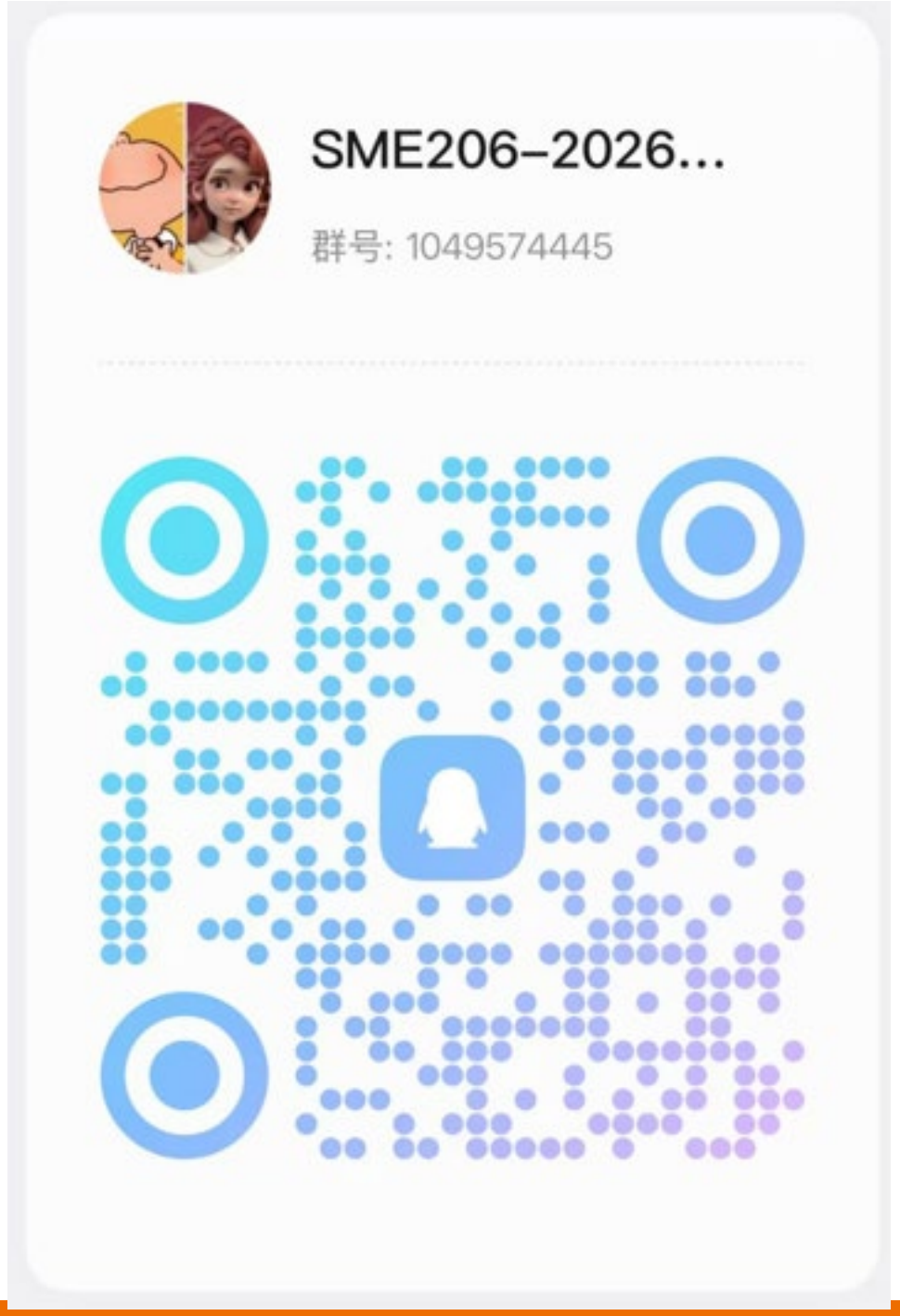

SME206-2026...
群号: 1049574445

SME206
Signals and Systems

0. Introduction

LI Jiamin

School of Microelectronics
Email: lijm3@sustech.edu.cn

Teaching Team

- **Lecturer**: Li Jiamin (李嘉敏)
  - Email: lijm3@sustech.edu.cn
  - Office: 智园崇文园区3号楼926
  - Office hour: 10am-12pm Mondays (**by appointment)**

Teaching Team

- **Teaching Assistants**
  - 朱英杰：12331334@mail.sustech.edu.cn
  - 刘芮志：12443039@mail.sustech.edu.cn
  - 吴若晗：12442040@mail.sustech.edu.cn
  - 杨明闯：12532791@mail.sustech.edu.cn

Time & Venue

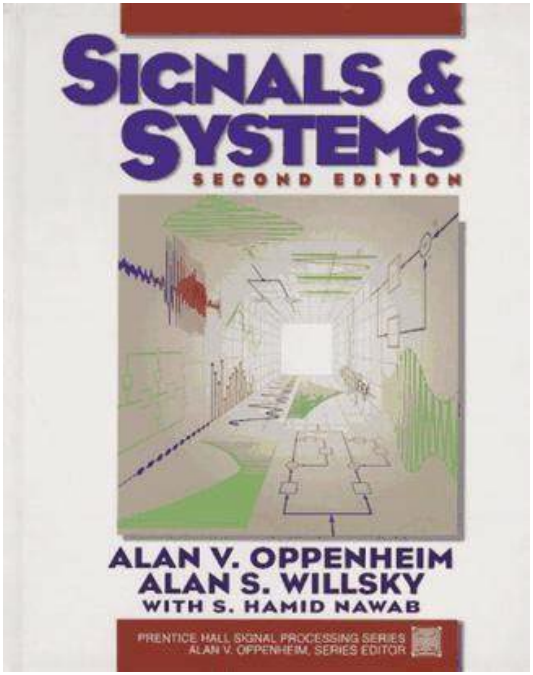
- **Lecture (32h)**
  - **Mon 19:00-20:50 @ 一教324**

- **Lab (32h)**
  - **Tue 14:00-15:50 @智华510机房**

Scope

"Signals and Systems", Oppenheim, Willsky and Nawab, 2nd Edition, 1997, Prentice-Hall

- **Chapter 1-8**

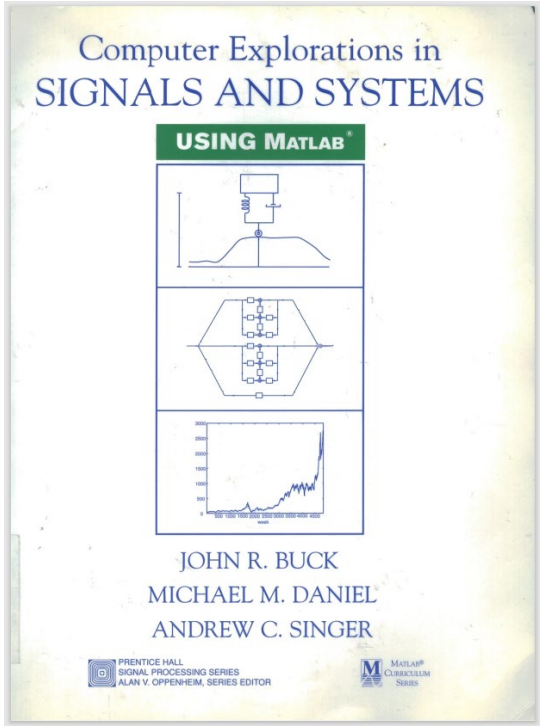


Textbook reading is highly recommended!

Scope

“Computer Explorations in Signals and Systems Using Matlab”, J. Buck, M.M. Daniel, A.C. Singer, 1997, Prentice-Hall

- **For lab practices**



Schedule

Week	Lecture	Assignment	Lab
1	Intro; Chap1 (Signal & System)		Lab1
2	Chap2 (LTI Systems)	HW1 due	Lab1
3	Chap2 (LTI Systems)		Lab2
4	Chap3 (Fourier Series of Periodic Signals)	HW2 due	Lab2
5			
6	Chap3 (Fourier Series of Periodic Signals)		Lab3
7	Chap4 (CT Fourier Transform)	HW3 due	Lab3
8	Chap4 (CT Fourier Transform)		Lab3

- Due date: 11:59pm Sunday

Schedule

Week	Lecture	Assignment	Lab
9	Mid-term Exam		Lab4
10	Chap5 (DT Fourier Transform)		Lab4
11	Chap6 (Time & Freq Characterization)	HW4 due	Project
12	Chap6 (Time & Freq Characterization)		Project
13	Chap7 (Sampling)		Project
14	Chap7 (Sampling)		Project
15	Chap8 (Communication System)		Project
16	Chap 8 & Final Review	HW5 due	Project

- Due date: 11:59pm Sunday. Project report due on week 17.

Assessment

- Attendance: 5%
- Assignment: 10%
- Lab: 10%
- Lab: Project: 15%
- Midterm Exam: 20%
- Final Exam: 40%

Pre-requisite Knowledge

- Calculus
- Linear algebra (basics)
- Complex number arithmetic
- Basic circuit theory
- Software programming background (for lab sessions)

- You should have learnt all these in Year 1.

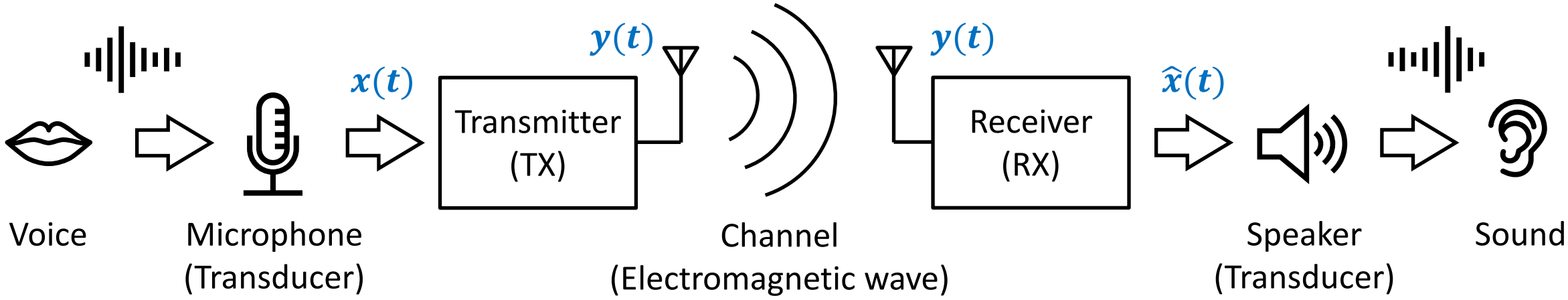
Terms you should be familiar with

- Complex number: magnitude and phase
- Complex conjugate
- Real & imaginary parts of complex number
  - relationship to magnitude & phase
- Exponential functions
- Sinusoids
- Frequencies – changing between Hz and radians/sec
- Period of a signal – relationship to frequency

Why study signals & systems?

- **What are the applications of signals & systems?**
  - **Communication**

Example: Making a phone call



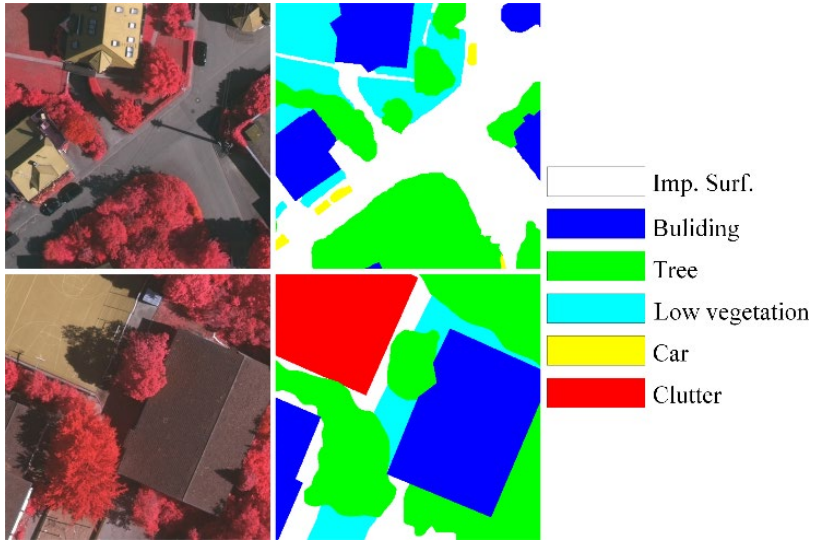
-- Signals? e.g. acoustic (speech, audio ..); electrical (voltage, current)

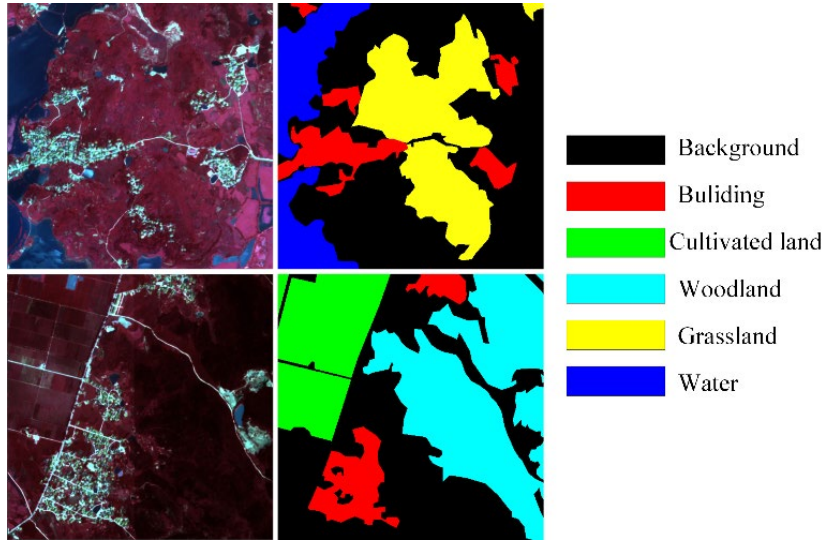
-- Systems? e.g. transmitter, channel, receiver, ...

Why study signals & systems?

- **What are the applications of signals & systems?**
  - **Signal processing**

Example: Image processing





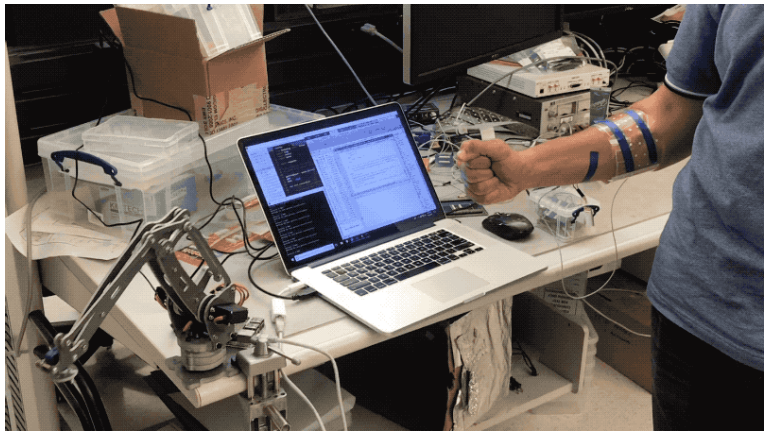
-- Signals? e.g. image, video, …

-- Systems? e.g. digital signal processor, …

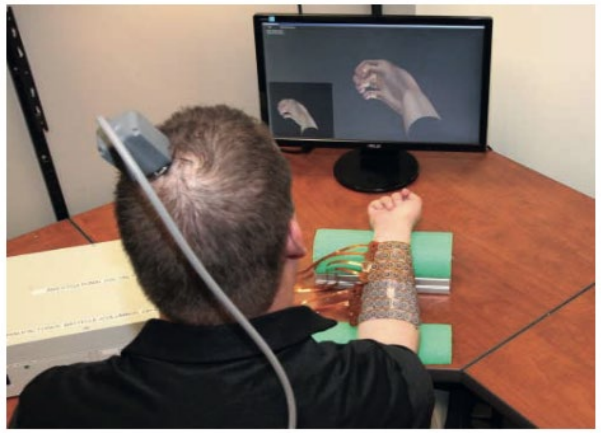
Why study signals & systems?

- **What are the applications of signals & systems?**
  - **Signal processing**

Example: Bio-signal Processing



FlexEMG, Berkeley, 2018



C.E. Bouton, Feinstein Institute & Ohio State Univ, 2016

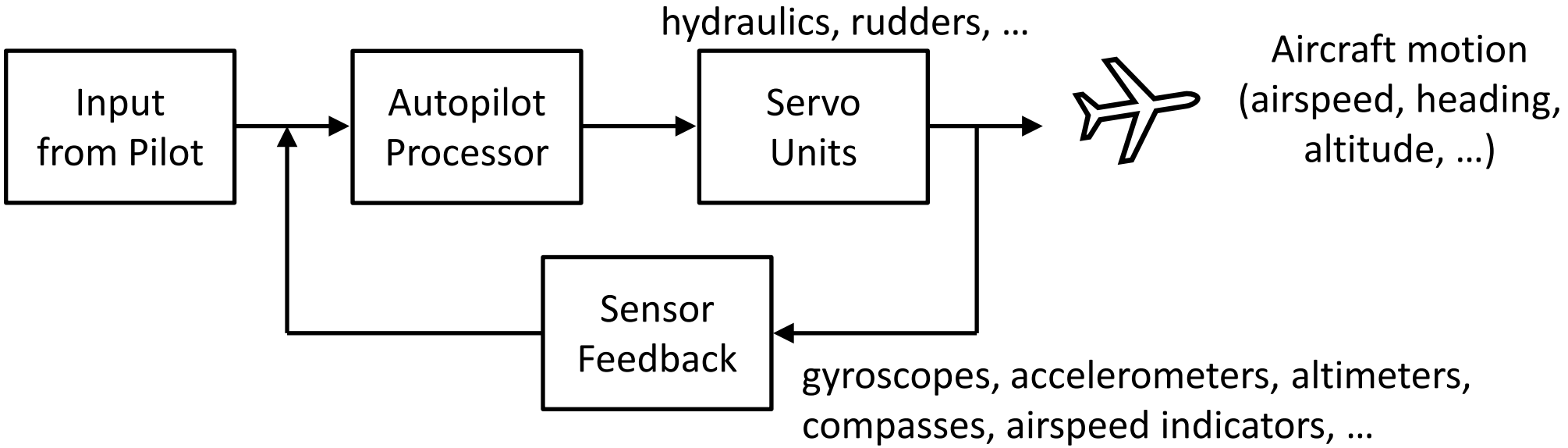
-- Signals? e.g. biological signals (EEG, ECG, EMG, ..)

-- Systems? e.g. acquisition, processor, stimulator, …

Why study signals & systems?

- **What are the applications of signals & systems?**
  - **Modelling & control**

Example: Auto-pilot control



-- Signals? e.g. pilot inputs, sensor data (airspeed, altitude, …)

-- Systems? e.g. autopilot, …

Why study signals & systems?

